

Optimization for Data Science

Lecture 1: Introduction to Optimization

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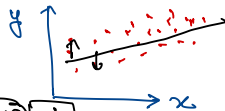
Outline

- 1 Course Overview
- 2 Grading Policy
- 3 What Is Optimization?
- 4 Examples in Data Science
- 5 Basic Mathematical Ideas in Optimization
- 6 Looking Ahead

Course Motivation

perfound yet
vanilla ML Algo!

Data Science + Machine Learning



Why Optimization for Data Science?

'TSP'

arguments:

align

- Many data science and machine learning methods boil down to parameter estimation that minimizes (or maximizes) an objective function (loss function, accuracy, etc.).
$$\min \sum (y_i - \hat{y}_i)^2$$
- A solid foundation in optimization helps in:
 - Efficiently training models (e.g., gradient-based methods for neural networks). $\hat{y} = \alpha x + \beta$
 - Handling large-scale problems (stochastic and distributed optimization).
 - Dealing with non-differentiable or black-box scenarios (heuristic/evolutionary algorithms).
- Understanding both theory and practice of optimization gives data scientists a sharper edge in solving real-world problems.

① SUR | SVM

② Lasso | Ridge | elastic
nets

α, β : parameters

unconstrained optimization!

→ constraint optm.

$\alpha < X$
$\beta < Y$
$\beta \geq Z$

Learning Objectives

By the end of the course, you should be able to:

- Understand the **mathematical foundations** of **optimization** (gradients, Hessians, convexity).
- Apply **classical** and **modern algorithms** (gradient descent, Newton's method, SGD, etc.).
- Incorporate heuristic methods such as **evolutionary algorithms** when gradient information is unavailable or the problem is highly nonconvex.
- Distinguish between **convex** and **nonconvex problems** and know typical solution strategies.
- Implement and **adapt optimization methods** in practical scenarios using real-world datasets.

Grading Policy

- Assignments/Tutorials: 30% ←
 - Exam 1: 30% ↔
 - Exam 2: 30% ↔ } "here" !
 - Class Participation: 10%
- 100%.

multiple short assignments
+
a few long assignments

General Definition

→ Convex Optimization
 → Momentum-based gradient Descent

→ prior knowledge!

Unconstrained Optimization Problem:

OR: (A) T-simplen
 → (B) Big-M ←

Constrained Optimization Problem:

$\min_{x \in \mathbb{R}^d} f(x).$
 {SUR | LASSO | Ridge}
 as Optim.

(C) → simplex ←
 (D) → Dual simplex ←

$\min_{x \in \mathcal{X}} f(x)$ subject to constraints on x .

→ Gradient Descent ↔
 → Adam Descent ↔

Typical Data Science Context:

→ Dynamic programming
 → SA | GA | PSO | ACO etc

→ Mathematical side!

- x often represents the parameters of a statistical or machine learning model.
- $f(x)$ could be a loss function or error metric we want to minimize.

(A) → swarm optimization



→ Evolutionary
 → Dual & Dual-simplex
 → Backpropagation!

→ Integer prog.
 + Quadratic

General Definition

2-D LPP: Graphs / graphical methods!!

$$\underline{\vec{x}} = \{ \underline{x_1}, \underline{x_2}, \dots, \underline{x_n} \}$$

Unconstrained Optimization Problem:

Integer $\rightarrow \mathbb{Z}$ $\mathbb{R}^d$: Real

$$\min_{\vec{x} \in \mathbb{R}^d} f(\vec{x}).$$

$\vec{x}$, x $\rightarrow$ single variable

vector of decision variables...

Constrained Optimization Problem:

$\min_{\vec{x} \in \mathcal{X}} f(\vec{x})$ subject to constraints on $\vec{x}$.

$$\min_{x, y, z \in \mathbb{R}} x^2 + y^2 + z^2$$

Typical Data Science Context:

$\times$ Ideal

Design: min. Cost!
 $\left\{ \begin{array}{l} \rightarrow \text{Resource constraints!} \\ \rightarrow \text{Money constraints!} \end{array} \right\}$

- $\vec{x}$ often represents the parameters of a statistical or machine learning model.
- $f(\vec{x})$ could be a loss function or error metric we want to minimize.

$$x \geq y \geq z \geq 0$$



Pareto
Surface

multiobjective programming
with constraints
"tradeoff"

$$\vec{x} \geq x, y, z$$

$$f(\vec{x}) = \underline{x^2 + y^2 + z^2} \quad \text{Generalized!}$$

Linear Regression

w = parameter

Ordinary Least Squares:

model!

$$\min_w \sum_{i=1}^n (y_i - \underline{w}^T \underline{x}_i)^2.$$

- Convex problem, can be solved via normal equations or gradient-based methods.
- Fundamental building block for many supervised learning tasks.

Logistic Regression

classification ...



Binary Classification:

$$\min_{\mathbf{w}} \sum_{i=1}^n [-y_i \ln(\sigma(\mathbf{w}^T \mathbf{x}_i)) - (1 - y_i) \ln(1 - \sigma(\mathbf{w}^T \mathbf{x}_i))],$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function.

- Also convex in $\mathbf{w}$, but no closed-form solution.
- Typically solved iteratively (gradient descent, Newton's method, etc.).

Evolutionary Algorithms

Ants:

Birds:

Genes:

Metals:

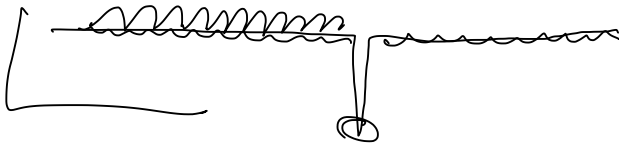
Don't see the math



Heuristic / Metaheuristic Approaches:

- Genetic Algorithms, Particle Swarm Optimization, Differential Evolution, etc.
- Particularly useful when:
 - The objective function is non-differentiable or black-box.
 - We aim for a global search over highly nonconvex landscapes.
 - Gradient information is costly or impossible to obtain.
- Often used in hyperparameter tuning, feature selection, and other discrete or complex optimization tasks within data science.

simulated
Annealing!



$$\max \equiv -\min$$

Local vs. Global Minima (Unconstrained):

- Local Minimum: A point $\mathbf{x}^*$ is a local min if there exists a neighborhood around $\mathbf{x}^*$ such that

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \text{for all } \mathbf{x} \text{ in that neighborhood.}$$

- Global Minimum: A point $\mathbf{x}^*$ is a global min if

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \text{for all } \mathbf{x} \text{ in the entire domain.}$$

Why This Matters:

- In data science, we often settle for local minima when f is nonconvex (like neural nets).
- If f is convex, local minima coincide with the global minimum, simplifying the search.

Stationary Points and Gradients

Unconstrained Setting:

- A stationary point $\mathbf{x}^*$ is one where

gradient
↓
 $\nabla f(\mathbf{x}^*) = \mathbf{0}$.

$\frac{\partial f(\vec{x})}{\partial x}$
 $f(\vec{x}) = [x^2 + y^2 + z^2]$

- For a **local minimum**, we often want $\nabla^2 f(\mathbf{x}^*)$ (the Hessian) to be *positive semidefinite*.
- If f is nonconvex, stationary points might be local maxima or saddle points, so checking the Hessian can help classify them.

Relevance:

- Many data science losses are smooth enough to have a gradient.
- If the function is convex, any stationary point is also a global min.

$\nabla f(\vec{x}) = \begin{Bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{Bmatrix}$

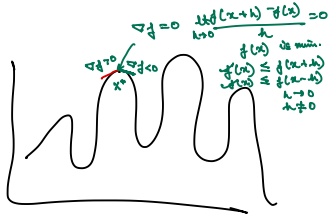
$$\frac{\partial f}{\partial x} = 0; \quad \frac{\partial f}{\partial y} = 0; \quad \frac{\partial f}{\partial z} = 0$$

Solution

necessary:

necessary condition: maxima/minima
(mathematically)

"Taylor expansion"



$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = 0$$

$f(x)$ is min.

$$\frac{f(x)}{f(x+h)} \leq \frac{f(x-h)}{f(x-h)}$$

$h \rightarrow 0$
 $h \neq 0$

$$\nabla f = 0$$



$\nabla f = 0$ does not guarantee ^{local} maxima or minima!!

saddle points? $\Rightarrow$

Given that it is stationary point; what is sufficient condition?

higher even derivative $\frac{\nabla^2 f}{\partial x^2} > 0$: minima!
why???

Constrained Problems (High-Level Note)

Constrained Optimization:

$$\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}), \quad \text{with constraints on } \mathbf{x}.$$

- Cannot simply set $\nabla f = 0$; the feasible region might restrict $\mathbf{x}$.
- Future lectures will introduce **Lagrange multipliers** (equality constraints) and **KKT conditions** (inequalities).

Examples in DS:

- $\|\mathbf{w}\| \leq R$ for interpretability or resource bounding.
- $\|\mathbf{w}\|_1 \leq \lambda$ (LASSO constraint) or $\|\mathbf{w}\|_1$ as a penalty.
- Nonnegativity in matrix factorization or bounding parameters in resource allocation.

- **Mathematical Preliminaries (Lecture 2):** Gradients, Hessians, vector calculus, convexity.
- **Classical Optimization (Lectures 3-4):** Line search methods, quasi-Newton, etc.
- **Constrained Optimization (Lectures 5-6):** Lagrange multipliers, KKT.
- **Linear Programming (Lectures 7-10):** Simplex, primal-dual, interior-point mention.
- **Evolutionary and Heuristic Methods (Lectures 11-13):** Genetic algorithms, advanced operators.
- **Multiobjective Optimization (Lectures 14-15):** Pareto approaches, evolutionary multiobjective.
- **ML/DS Applications (Lectures 16-20):** Regression, LASSO, SVR, neural nets, custom loss.

Key Takeaways

- Optimization underpins most data science and machine learning tasks.
- We start simple (unconstrained) and build up to advanced methods (constrained, evolutionary, multiobjective).
- Course approach: theory + practical angles, ensuring you can implement these techniques in real scenarios.

Thank You!

Questions or Comments?